

Reference Sheet

Not everything on this reference sheet may be needed.

String Class API

```
public class String {  
    /** Returns the character at the specified index (position) */  
    public char charAt(int index) { ... }  
  
    /** Returns the substring beginning at index from and ending at index to - 1 */  
    public String substring(int beginIndex, int endIndex) { ... }  
  
    /** Returns the length of this string. */  
    public int length() { ... }  
  
    /** Returns true if this is equal to other; returns false otherwise */  
    public boolean equals(String other)  
}
```

Random Class API

```
public class Random {  
    /** Creates a new random number generator. */  
    public Random() { ... }  
  
    /** Returns a pseudorandom, uniformly distributed int value between 0 (inclusive) and the  
    specified value (exclusive), drawn from this random number generator's sequence. */  
    public int nextInt(int bound) { ... }  
}
```

Set Interface

```
public interface Set<T> ... { ...  
    /** Adds the specified element to this set if it is not already present */  
    boolean add(T obj)  
  
    /** Returns true if this set contains the specified element */  
    boolean contains(T obj)  
  
    /** Compares the specified object with this set for equality. */  
    boolean equals(Object obj)  
  
    /** Removes the specified element from this set if it is present */  
    boolean remove(T obj)  
}
```

Map Interface

```
public interface Map<K, V> ... { ...  
    /** Returns true if this map contains a mapping for the specified key */  
    boolean containsKey(K key)  
  
    /** Returns true if this map maps one or more keys to the specified value. */  
    boolean containsValue(V value)  
  
    /** Returns the value to which the specified key is mapped, or null if this map contains no  
    mapping for the key. */  
    V get(K key)  
  
    /** Associates the specified value with the specified key in this map */  
    V getOrDefault(K key, V defaultValue)  
  
    /** Returns the value to which the specified key is mapped, or null if this map contains no  
    mapping for the key. */  
    V put(K key, V value)  
}
```

List Interface

```
public interface List<T> ... { ...  
    /** Returns the number of elements in the list */  
    int size()  
  
    /** Appends obj to end of list; returns true */  
    boolean add(T obj)  
  
    /** Returns the element at position index in the list */  
    T get(int index)  
  
    /** Replaces the element at position index with obj; returns the element formerly at position  
    index */  
    T set(int index, T obj)  
  
    /** Removes element from position index, moving elements at position index + 1 and higher to the  
    left (subtracts 1 from their indices) and subtracts 1 from size; returns the element formerly at  
    position index */  
    T remove(int index)  
}
```